Qafila gross margin

Project explanation and verified results

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The business question

Revenue alone does not tell Qafila what it earns after paying suppliers. This pipeline combines order lines with weekly supplier quotes to calculate margin by ISO week, city and product category.

From source files to published tables

Stage	Responsibility
Python / DuckDB raw	Read application tables from SQLite and supplier CSVs as received. Keep each supplier filename and row number.
dbt staging	Trim SKUs, parse mixed cost formats, convert UTC order times to Riyadh dates and calculate line revenue.
dbt quality / silver / gold	Quarantine invalid supplier rows, build cost history and calculate weekly margin.
Apache Iceberg	Publish three tables. Skip unchanged results and retain older snapshots when data changes.

Published table grain

Table	One row per
silver.product_cost_history	SKU and cost version
gold.weekly_margin	ISO week, city and category
quality.price_quarantine	Rejected supplier row

Source acknowledgement

Adapted from the SDAIA Academy assignment and the supplied examples. The main implementation reference is reem7628-blip/project; the dee2002al-c and ariffalshehri examples were also reviewed. See ACKNOWLEDGEMENTS.md for the source revisions.

Rules and evidence

Supplier cost history

Unknown SKUs are rejected first. Other rows are rejected when their cost is missing, not finite, zero or negative. Accepted quotes are deduplicated by SKU and effective date, with the last filename winning. A new SCD2 version starts only when the rounded cost changes. Its end date is the next version's start date, exclusive.

Costing order lines

Count delivered orders with positive quantities and Riyadh order dates on or after 2026-08-31. Join each line to the cost valid on its order date. Lines without a matching cost remain in revenue and are reported as uncosted. Margin uses only the revenue of costed lines.

```
line revenue = round(quantity * unit_price - discount, 2)
line cost    = quantity * unit_cost_sar
margin       = costed_revenue_sar - cost_sar
```

Measured results: seed 2026

Measure	Before correction	After correction
Cost versions	633	637
Weekly margin rows	165	165
Quarantined rows	7	7
Total cost / SAR	1,654,199.98	1,651,701.06
Total margin / SAR	801,016.88	803,515.80

Verification coverage

Seeds 2026 and 11 each passed 26 dbt tests and 22 contract checks. All three outputs matched the independent Python reference before and after correction. Re-runs preserved rows, table identities and snapshots. Every original snapshot remained readable after correction. Three supplier-ingestion regression tests also passed.

Run tools/verify_pipeline.py to reproduce the integration checks. The measured snapshot IDs are in docs/verification.json. Temporary warehouses from that verification are removed on exit.